Production deployment validation matrix

Production deployment validation matrix - 2.7.x upgrade

Status: **NOT READY FOR PRODUCTION** (see the gate rule).

Environment: development host `172.16.16.44` (`ocrv2`), release `2.7.0`, validated 12 November 2026.

Gate rule

Promote to production **only when every required row below is PASS** and the release owner has signed the bottom of this file. A single FAIL or NOT RUN blocks promotion. Optional rows do not block, but must be listed in the release notes.

Matrix

ID	Gate	Required	Status	Evidence
B1	Backend unit/integration tests	Yes	PASS	`pytest` : 290 passed, 179 skipped locally; CI runs the DB-backed set on PostgreSQL 17
B2	Frontend lint	Yes	PASS	0 errors, 132 warnings
B3	Frontend production build	Yes	PASS	`vite build` succeeds
B4	Lab validation	Yes	PASS	`validate-lab.py --all` : 169 labs, 0 errors, 0 warnings
B5	Supply-chain lock	Yes	PASS	42 files, 169 lab hashes match
B6	Licence compliance	Yes	PASS	`check-licences.py` : OK
B7	SPA security headers / CSP hash	Yes	PASS	`check-nginx-security-headers.py` , `check-csp-hash.py`
B8	External images in learner content	Yes	PASS	`check-external-images.py` : none
B9	Documentation links and assets	Yes	PASS	577 Markdown files, no broken links; assets PASS
B10	Compose config and image builds	Yes	PASS	Backend and frontend images build; compose config valid
B11	Alembic head	Yes	PASS	Dev DB at `0014_performance_indexes` ; startup upgrade clean

ID	Gate	Required	Status	Evidence
R1	Health and version on dev	Yes	PASS	`/health` 200, `/api/version` 2.7.0
R2	Connection-pool burst	Yes	PASS	36 concurrent dashboard/admin requests in 2s; 11/20 pool connections; no QueuePool errors
R3	Scheduler tick responsiveness	Yes	PASS	16/16 health samples at 200 through the 120s tick
R4	Software integrity endpoint	Yes	PASS	169/169 labs match; 0 mismatched/missing/extra; 0.2s
R5	CyberChef panel	Yes	PASS	`/tools/cyberchef/` 200, assets 200, inline-script hashes match CSP
R6	High-contrast theme	Yes	PASS	Theme cycle dark/light/contrast; contrast tokens present in the built bundle
R7	Mobile shell (student)	Yes	PASS	390x844: dashboard, exercises, profile, rangebox; 0 overflow, 0 console errors, tab bar present
R8	Mobile shell (admin/instructor)	Yes	PASS	390x844: dashboard, admin, instructor; notices present; 0 overflow/errors
R9	New OT lab (S7comm)	Yes	PASS	Seeded id 187, image prebuilt, DB1 read returns the flag
R10	Image upload hardening	Yes	PASS	7 tests: PNG accepted/re-encoded; SVG, HTML, mismatch, oversized rejected
D1	FK orphan scan (all FKs)	Yes	PASS	No orphan rows reported
D2	Nulls in critical columns	Yes	PASS	users/labs/levels/sessions/completions/settings: 0
D3	Duplicate usernames/emails/slugs	Yes	PASS	0
D4	Numeric sanity	Yes	PASS	No negative durations/counters, no future timestamps
D5	Ledger attribution	No	WARN	6,928 of 7,785 rows unattributed (seed/CLI/system writes); review queue noisy
D6	Stale lab rows	No	PASS	178 DB labs = 169 repo + 9 intentional `demotest-*`
P1	Backup restore drill	Yes	NOT RUN	No restore from a real snapshot in this cycle
P2	Upgrade rehearsal 2.6 -> 2.7	Yes	NOT RUN	Dev was already on 2.7.x
P3	Rollback rehearsal	Yes	NOT RUN	No documented rollback test this cycle
P4	Signed release tag	Yes	NOT RUN	`v2.7.0` remains the documented unsigned exception; new tag not cut
P5	Production-shaped host rehearsal	Yes	NOT RUN	No staging host identical to production
P6	Email delivery	Yes	NOT RUN	No SMTP transaction tested
P7	WireGuard / Peer Manager	Yes	NOT RUN	No VPN client enrolled this cycle

ID	Gate	Required	Status	Evidence
P8	RangeBox desktop launch	Yes	NOT RUN	No interactive desktop session this cycle
P9	Exhaustive button/flow E2E	No	PARTIAL	Route-level Playwright smoke passed; not every button
P10	Dependency/image digest pinning	No	PARTIAL	Base images remain tag-pinned; digests reported at runtime

Result

- Required PASS: 25
- Required FAIL: 0
- Required NOT RUN: 8 (P1-P8)
- Warnings: 1 (D5)
- Completion: **25 of 33 required gates (76%)**

Decision: DO NOT PROMOTE. The application is healthy on dev, but the production-readiness gates for restore, upgrade/rollback rehearsal, signed release, and the infrastructure integrations are not complete.

Exit criteria for the next attempt

1. Run P1-P8 on a production-shaped host and record evidence in this file.
2. Cut and sign the release tag; update `docs/RELEASES.md`.
3. Either fix D5 (set actor context for seed/discovery/scheduler writes) or record an accepted-risk note with a retention policy for the ledger.
1. Re-run B1-B10 and R1-R10 on the release commit.
2. Only then change this file to `READY FOR PRODUCTION` and sign below.

Sign-off

Role	Name	Decision	Date
Release owner			
Data analyst		Conditional (see D5)	2026-11-12
Security reviewer		Pending P4/P10	2026-11-12